15. Find $P \cup Q$.
    A. $\{\ \}$
    B. $\{2\}$
    C. $\{0, 2, 4, 6, 8, 10\}$
    D. $\{2, 4, 6, 8, 10, 12\}$
16. On dividing $6\sqrt{27}$ by $2\sqrt{3}$, we obtain ———.
    A. $3\sqrt{9}$
    B. 6
    C. 9
    D. none of these
17. A rational number between 1/7 and 2/7 is ———.
    A. $\frac{2}{21}$
    B. $\frac{5}{21}$
    C. $\frac{1}{14}$
    D. $\frac{5}{14}$
18. The rational number $0.\overline{3}$ can also be written as _____.
    A. 0.3
    B. $\frac{3}{10}$
    C. 0.33
    D. $\frac{1}{3}$
19. A rational number equivalent to 5/7 is ———.
    A. $\frac{15}{17}$
    B. $\frac{25}{27}$
    C. $\frac{10}{14}$
    D. $\frac{10}{27}$
20. If two matrices A and B can be multiplied to obtain the matrix AB, then _____.
    A. both A and B are rectangular.
    B. both A and B have the same order.
    C. the number of columns of A is equal to the number of rows of B.
    D. the number of rows of A is equal to the number of columns of B.

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UNIVERSITY OF CAPE COAST
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SEMESTER I QUIZ I, 2019/2020
DIPLOMA IN MATHEMATICS AND SCIENCE EDUCATION
DME / DSE YR

MAT 101: ALGEBRA AND TRIGONOMETRY

EXAM CODE: 02

NAME: .....

REGISTERED NO:..... **CANDIDATE'S SIGNATURE:**.....

STUDY CENTRE:.....**TIME ALLOWED: 40 MINUTES**

INSTRUCTION: ***Answer all questions by circling the correct option in ink.***

1. Given the sets A, B and C, which one of the following set identities is false?
    A. $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
    B. $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
    C. $(A \cup B) \times C = (A \times C) \cap (B \times C)$.
    D. $(A \cap B) \times C = (A \times C) \cap (B \times C)$.
2. If we include "0" in the set of natural numbers, the resulting set is said to be the set of ———.
    A. natural numbers
    B. whole numbers
    C. prime numbers
    D. odd numbers.
3. $(16)^{3/4}$ is equal to ———.
    A. 2
    B. 4
    C. 8
    D. 16
4. To rationalize the denominator of the expression $\frac{1}{\sqrt{12}}$, we multiply and divide by _____.
    A. $\sqrt{2}$.
    B. $\sqrt{3}$.
    C. $\frac{1}{\sqrt{12}}$.
    D. 12.

5. Simplifying: $(\sqrt{5}+\sqrt{7})^2$.
   A. $\sqrt{5}+\sqrt{7}$
   B. $\sqrt{35}$
   C. 12
   D. $12+2\sqrt{35}$

6. Let $|A|$ be the determinant of the matrix A. If $|A|=0$, then A _____.
   A. zero matrix
   B. singular matrix
   C. non-singular matrix.
   D. 0.

7. Find the inverse of the matrix $A=\begin{bmatrix}-5 & 4\\ 0 & 4\end{bmatrix}$, if it exists.
   A. $\begin{bmatrix}-\frac{1}{5} & -\frac{1}{5}\\ 0 & \frac{1}{4}\end{bmatrix}$
   B. $\begin{bmatrix}-\frac{1}{5} & \frac{1}{5}\\ 0 & \frac{1}{4}\end{bmatrix}$
   C. $\begin{bmatrix}\frac{1}{4} & \frac{1}{5}\\ 0 & -\frac{1}{5}\end{bmatrix}$
   D. $\begin{bmatrix}0 & \frac{1}{4}\\ -\frac{1}{5} & \frac{1}{5}\end{bmatrix}$

8. Calculate the determinant of the matrix $B=\begin{bmatrix}4 & 2 & 7\\ 9 & 3 & 5\\ 7 & 9 & 4\end{bmatrix}$
   A. -286
   B. 146
   C. 286
   D. 1084

9. Suppose that $U=\begin{bmatrix}1 & -1\\ 0 & -3\\ 5 & 2\end{bmatrix}$ and $V=\begin{bmatrix}-2 & 1\\ 5 & 4\\ 0 & -7\end{bmatrix}$. Evaluate $7U+6V$.
   A. $\begin{bmatrix}19 & -13\\ 30 & 3\\ -35 & 56\end{bmatrix}$
   B. $\begin{bmatrix}-5 & -1\\ 0 & 3\\ 0 & -28\end{bmatrix}$
   C. $\begin{bmatrix}-5 & -1\\ -30 & -45\\ 35 & 56\end{bmatrix}$
   D. $\begin{bmatrix}-5 & -1\\ 30 & 3\\ 35 & -28\end{bmatrix}$

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10. Solve the system of equations $\begin{cases}-3x+10y=3\\ x-3y=-3\end{cdot}$
    A. $\begin{bmatrix}21\\ 6\end{bmatrix}$
    B. $\begin{bmatrix}-6\\ -21\end{bmatrix}$
    C. $\begin{bmatrix}-21\\ -6\end{bmatrix}$
    D. No solution.

11. What is the definition of the union of two sets A and B?
    A. The set of all elements that are in both set A and in set B.
    B. The set of elements that are either in set A or in set B.
    C. The set of elements that are in set A, but not in set B.
    D. The set of elements that are in set B, but not in set A.

12. Two groups of students, group A and group B surveyed their campus for student satisfaction. Group A surveyed students from the math, science, and philosophy departments. Group B surveyed students from the language, business, and philosophy departments. What could be considered the universal set in this scenario?
    A. The set of all students on campus.
    B. The set of all students in the philosophy department.
    C. The set of all students surveyed by group A.
    D. The set of all students surveyed by group B.

13. Which of the following set is equivalent to the set {c, r, e, a, t, e}?
    A. {4}
    B. {5}
    C. {L, O, V, E}
    D. {f, a, i, t, h}

Suppose that the set $U=\{x:0<x\leq 12, x\in \text{Integers}\}$ *and* P *and* Q *are subsets of U given by* $P=\{x:x \text{ is divisible by } 2\}$ *and* $Q=\{x:x \text{ is a prime}\}$. ***Use the information to answer Questions 14 and 15.***

14. Find $P\cap Q$.
    A. {0, 2, 4, 6, 8, 10}
    B. {2, 4, 6, 8, 10, 12}
    C. {2}
    D. { }